



Politecnico di Torino

Qubit Electronics

Lab 5

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Lab submission instructions

Create a folder with the following name:

`gr<nn>_lab5`

where n corresponds to the group number.

In the root of the lab folder, insert the .pdf with the report. *The report's length must be limited to 10 pages.*

At the end of the lab session, create a .zip archive starting from the root folder and name it

`gr<nn>_lab5.zip`

The created archive must be uploaded by one team member on Portale della Didattica, section Elaborati.

1 Introduction

During this laboratory session, we will focus on analyzing a real double quantum dot device, where confinement is achieved by exploiting Silicon-Silicon Dioxide (Si-SiO₂) materials, by solving the Schrodinger equation.

2 Modify the code

Modify the code of the previous laboratory in order to solve the Schrodinger equation for the device under investigation. The constraints are:

- Number of energy levels to consider is equal to 5
- The tolerance for the convergence is equal to $1e-9$

Question: Report the modified code commenting your modifications.

3 Results

As done in the first lab on QTCAD, perform all analysis on the generated output files.

Question: Report all the results and also make some comments on them. In particular:

- What do the results represent?
- Does the device work as expected?
- In what bonding configuration are the two dots located?